Solution



(1) (25 pts.) Find the following for the matrix: $A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 3 \\ 0 & 1 & -1 \end{bmatrix}$. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ -1 & 3 & -1 \end{bmatrix}$

(a) Find the determinant. (b) True or False: $A$ is invertible. (c) Find the characteristic polynomial. (d) Find the eigenvalues. (e) For each eigenvalue, find a basis for the eigenspace.

(c) 5 char poly $A = (1-\lambda)((1-\lambda)(-1-\lambda)-3) = (1-\lambda)(\lambda^2-4) = (1-\lambda)(\lambda-2)(\lambda+2)$

(expand c1) → (a) 3 $\det A = 1(-1-3) = -4$ 2 (b) True

3 (d) $\lambda = 1, 2, -2$

(e) $\lambda_1 = 1$ $A - 1 = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 3 \\ 0 & 1 & -2 \end{bmatrix} \to \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $v_1 = a\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

12

$\lambda_2 = 2$ $A - 2 = \begin{bmatrix} -1 & 0 & -1 \\ 0 & -1 & 3 \\ 0 & 1 & -3 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$ $v_2 = c\begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix}$

$\lambda_3 = -2$ $A + 2 = \begin{bmatrix} 3 & 0 & -1 \\ 0 & 3 & 3 \\ 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & -1/3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ $v_3 = c\begin{bmatrix} 1/3 \\ -1 \\ 1 \end{bmatrix}$

---

Version 3G (c) char poly $= (1-\lambda)(\lambda-2)(\lambda+2)$

(a) $|A| = -4$ (b) True (d) $1, 2, -2$

(e) $\lambda_1 = 1$ $v_1 = \begin{bmatrix} 1 \\ 1/3 \\ 0 \end{bmatrix}$

$\lambda_2 = 2$ $v_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$

$\lambda_3 = -2$ $v_3 = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}$

(2) (25 pts.) Answer the following for the vectors:

$$\mathbf{w}_1 = \begin{bmatrix}0\\0\\2\\1\end{bmatrix}, \mathbf{w}_2 = \begin{bmatrix}1\\1\\0\\1\end{bmatrix}, \mathbf{w}_3 = \begin{bmatrix}1\\1\\4\\3\end{bmatrix}, \mathbf{w}_4 = \begin{bmatrix}0\\-1\\3\\1\end{bmatrix}, \mathbf{v} = \begin{bmatrix}1\\-1\\4\\2\end{bmatrix}.$$

Let $W$ be the subspace of $\mathbb{R}^4$ spanned by the set of vectors: $\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3, \mathbf{w}_4\}$. (a) Find a basis for $W$. (b) What is the dimension of $W$? (c) Is $\mathbf{v}$ in $W$? If true, write $\mathbf{v}$ as a linear combination of the basis given in Part (a). (d) Extend the basis for $W$ given in Part (a) to a basis for $\mathbb{R}^4$.

$$[w_1\ w_2\ w_3\ w_4\ v] = \begin{array}{c} -1 \\ \\ \\ -2 \end{array}\begin{bmatrix}0&1&1&0&1\\0&1&1&-1&-1\\2&0&4&3&4\\1&1&3&1&2\end{bmatrix} \to \begin{bmatrix}1&1&3&1&2\\0&1&1&0&1\\0&0&0&-1&-2\\0&-2&-2&1&0\end{bmatrix}\begin{array}{c}2\\ \\ \\ \\ \end{array} \to \begin{bmatrix}1&0&2&1&1\\0&1&1&0&1\\0&0&0&-1&-2\\0&0&0&1&2\end{bmatrix}$$

$$\to \left[\begin{array}{cccc|c}1&0&2&0&-1\\0&1&1&0&1\\0&0&0&1&2\\0&0&0&0&0\end{array}\right] \leftarrow \text{RREF}$$

10

2

(a) basis $W = \{w_1, w_2, w_4\}$ (b) 3 (c) Yes

3

$$v = -\begin{pmatrix}0\\0\\2\\1\end{pmatrix} + \begin{pmatrix}1\\1\\0\\1\end{pmatrix} + 2\begin{pmatrix}0\\-1\\3\\1\end{pmatrix}$$

5

(d) adjoin $e_1$ $[w_1\ w_2\ w_4\ e_1] = \begin{array}{c} \\ \\ \\ -2 \end{array}\begin{bmatrix}0&1&0&1\\0&1&-1&0\\2&0&3&0\\1&1&1&0\end{bmatrix} \to \begin{bmatrix}1&1&1&0\\0&1&0&1\\0&0&-1&-1\\0&-2&1&0\end{bmatrix}\begin{array}{c}2\\ \\ \\ \\ \end{array}$

5

$$\to \begin{bmatrix}1&1&1&0\\0&1&0&1\\0&0&1&1\\0&0&1&2\end{bmatrix} \to \begin{bmatrix}1&1&1&0\\0&1&0&1\\0&0&1&1\\0&0&0&1\end{bmatrix} \leftarrow \text{echelon form rank} = 4$$

$\{w_1, w_2, w_4, e_1\}$ is a basis for $\mathbb{R}^4$.

---

Version 3G $[w_1\ w_2\ w_3\ w_4\ v] \xrightarrow[\text{RREF}]{}$ same as above

(a) (b) (c) (d) same as above

(3) (25 pts.) Answer the following for the matrix: $A = \begin{bmatrix} -1 & 1 \\ 4 & -1 \end{bmatrix}$.

(a) Find the determinant, rank and nullity of $A$.
(b) True or False: $A$ is invertible. If true, find $A^{-1}$.
(c) Find the characteristic polynomial of $A$.
(d) Find the eigenvalues of $A$.
(e) For each eigenvalue, find a basis for the eigenspace.

(a) $\det = \begin{vmatrix} -1 & 1 \\ 4 & -1 \end{vmatrix} = 1-4 = -3$ (2), rank $A = 2$ (2), Null $A = 0$ (1)

5 (b) True $A^{-1} = -\frac{1}{3}\begin{bmatrix} -1 & -1 \\ -4 & -1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$

2 (c) char poly $= \begin{vmatrix} -1-\lambda & 1 \\ 4 & -1-\lambda \end{vmatrix} = (\lambda+1)^2 - 4 = \lambda^2+2\lambda-3 = (\lambda+3)(\lambda-1)$

1 (d) $\lambda_1, \lambda_2 = -3, 1$

(e) $\lambda_1 = -3$ $A+3 = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \to \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$ $\lambda_1 = b\begin{pmatrix} +1 \\ -2 \end{pmatrix}$

6 $\lambda_2 = 1$ $A-1 = \begin{bmatrix} -2 & 1 \\ 4 & -2 \end{bmatrix} \to \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix}$ $\lambda_2 = b\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

Version 3G

(a) $\begin{vmatrix} -1 & 4 \\ 1 & -1 \end{vmatrix} = -3$ (b) $A^{-1} = -\frac{1}{3}\begin{bmatrix} -1 & -4 \\ -1 & -1 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}$

(c) $\begin{vmatrix} -1-\lambda & 4 \\ 1 & -1-\lambda \end{vmatrix} = (\lambda+1)^2 - 4 = \lambda^2+2\lambda-3 = (\lambda-1)(\lambda+3)$

(d) $\lambda_1, \lambda_2 = -3, 1$

(e) $\lambda_1 = -3$ $A+3 = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$ $v_1 = b\begin{pmatrix} -2 \\ 1 \end{pmatrix}$

$\lambda_1 = 1$ $A-1 = \begin{bmatrix} -2 & 4 \\ 1 & -2 \end{bmatrix} \to \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}$ $v_2 = b\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

(4) (25 pts.) Answer the following for the set: $W = \left\{ \begin{bmatrix} b+c \\ -a+b \\ a+c \end{bmatrix} : a, b, c \in \mathbb{R} \right\}$.

(a) Find a basis for $W$. (b) What is the dimension of $W$? (c) True or False: $W$ is equal to $\mathbb{R}^3$. (d) For each of the vectors

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 3 \\ 4 \\ -1 \end{bmatrix},$$

determine whether it is in $W$. If yes, write it as a linear combination of the basis vectors listed in Part (a). (e) Extend the basis listed in Part (a) to a basis for $\mathbb{R}^3$.

$W$ = col space of $A = \begin{bmatrix} 0 & 1 & 1 \\ -1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ $\xrightarrow{\text{reduce}}$ $(A \mid u_1 u_2 u_3) = \left[\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 3 \\ -1 & 1 & 0 & 0 & 1 & 4 \\ 1 & 0 & 1 & 0 & 0 & -1 \end{array}\right]$

10

$R_1 \leftrightarrow R_3$, $R_2 + R_3$: $\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 1 & 0 & 3 \end{array}\right] \to$ $R_3 - R_2$: $\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & -1 \\ 0 & 1 & 1 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 & -1 & 0 \end{array}\right]$ RREF

3

(a) $\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ (b) dim $W = 2$ (c) False (d) $u_1$: No $u_2$: No

1 1 1

$u_3$: Yes : $u_3 = -\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} + 3\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

5

(e) 3 $\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ ← adjoin $u_1$

Version 3G $(A \mid u_1 u_2 u_3) = \left[\begin{array}{ccc|ccc} 0 & 1 & 1 & 0 & 1 & 1 \\ -1 & 1 & 0 & 1 & 0 & 4 \\ 1 & 0 & 1 & 0 & 0 & -3 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & -3 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 \end{array}\right] -1$

$\to \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & -3 \\ 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & -1 & 1 & 0 \end{array}\right] -1 \to \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 0 & 0 & -3 \\ 0 & 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & -1 & 0 \end{array}\right]$ RREF

(a) $\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ (b) 2 (c) False (d) $u_1$: No $u_2$: No $u_3 = -3\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

(e) adjoin $u_1$ or $u_2$